higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

MATERIAL TECHNOLOGY
NQF LEVEL 3

(6021013)

14 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 8 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Clearly cross out ALL work you do NOT want to be marked.
5. Start each section on a NEW page.
6. Use only BLUE or BLACK ink.
7. Write neatly and legibly.

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QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the correct answer and write only the letter (A–D) next to the question number (1.1.1–1.1.5) in the ANSWER BOOK.

1.1.1 The ability of a material to return to its original shape after a load has been removed from it:

A Malleability
B Elasticity
C Tenacity
D Compressive strength

1.1.2 The tendency of a metal to break or crumble to pieces when a sudden load is applied:

A Elasticity
B Brittleness
C Toughness
D Tensile strength

1.1.3 Ferrous metals consist mainly of:

A Aluminium
B Chrome
C Iron
D Manganese

1.1.4 The property of a material to be drawn out to a considerable length without breaking:

A Malleability
B Toughness
C Ductility
D Tenacity

1.1.5 The ability of a metal to resist crushing forces:

A Compressive strength
B Tensile strength
C Stress
D Strain

(5 × 1) (5)

1.2 Define the following physical properties of metals:

1.2.1 Density

1.2.2 Melting point

1.2.3 Thermal conductivity

(3 × 2) (6)

1.3 When the carbon content of steel is increased, it affects some of the mechanical properties of the steel. Briefly explain the effects that increased carbon content in steel has on the following mechanical properties of the steel:

1.3.1 Hardness

1.3.2 Tensile strength

1.3.3 Ductility

1.3.4 Machinability

(4 × 1) (4)

1.4 A puddling furnace is used for batch-type melting of bronze or brass where the charge in the furnace is not heated by direct contact with the fire.

1.4.1 Explain how the charge is heated in the furnace. (2)

1.4.2 Name TWO types of fuel used to fire a puddling furnace. (2)

1.4.3 The furnace is lined with firebricks on the inside.

Explain the function or purpose of the firebricks in detail. (3)

1.5 Molybdenum is an alloy element used to change the mechanical properties of steel.

State THREE (positive) changes that molybdenum, as an alloying element, can make to the mechanical properties of steel. (3)

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QUESTION 2

2.1 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (2.1.1–2.1.10) in the ANSWER BOOK.

2.1.1 Non-metals are shown on the right-hand side of the periodic table.

2.1.2 The symbol for Arsenic is Ar.

2.1.3 Non-metals are brittle and hence not malleable.

2.1.4 Non-metals are very strong.

2.1.5 Boron can be used in tools.

2.1.6 The symbol for Bromine is Br.

2.1.7 Pure diamonds are poor conductors of electricity.

2.1.8 Methane gas is known as a hydrocarbon.

2.1.9 Graphite can be used as a lubricant.

2.1.10 Phosphorous with the symbol Ph is a non-metallic element shown on the periodic table

(10 × 1) (10)

2.2 Plastics are widely used in industry today and are even sometimes used instead of metals.

2.2.1 Toughness is a property of plastics.

Name FOUR other properties of plastics. (4)

2.2.2 Name FOUR non-engine related applications of plastics in the automobile industry. (4)

2.3 Rubber has become a widely used substance in the industry and is well known for the manufacturing of tyres.

2.3.1 Name FOUR properties of rubber. (4)

2.3.2 Styrene-butadiene rubber is one example of a synthetic rubber.

Name THREE other types of synthetic rubber. (3)

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QUESTION 3

3.1 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–K) next to the question number (3.1.1–3.1.10) in the ANSWER BOOK.

COLUMN A		COLUMN B	
3.1.1	Glass-like waste product	A	blast furnace
3.1.2	Water that contains salt	B	cupola furnace
3.1.3	Used to join two metals together	C	impact test
3.1.4	Soak or dip into liquid to lower temperature	D	nitriding
		E	ore
3.1.5	Spoon-shaped vessel for transferring molten metal	F	scale
3.1.6	Converts pig iron to cast iron	G	quench
3.1.7	Hard, rough substance that forms on the surface of steel	H	brine
		I	ladle
3.1.8	Applying controlled blow to a surface	J	flux
3.1.9	Surface hardening of mild steel	K	slag
3.1.10	Minerals containing metals		

(10 × 1) (10)

3.2 The electric arc furnace is used to produce good quality alloy steels from scrap metal and operates in four stages. The FOUR stages are:

Stage 1: (3.2.1)

Stage 2: The carbon electrodes are lowered to the surface of the metal and melting starts. When the melting is complete, the slag floats to the top. Iron ore is added to remove carbon and phosphorous.

Stage 3: (3.2.2)

Stage 4: (3.2.3)

3.2.1 Explain Stage 1 of the electric arc furnace. (2)

3.2.2 Explain Stage 3 of the electric arc furnace. (2)

3.2.3 Explain Stage 4 of the electric arc furnace. (3)

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3.3 Heat treatment is a method used to alter the physical properties of materials.

3.3.1 Tempering is one example of a heat treatment method.

Name THREE other methods of heat treatment. (3)

3.3.2 Briefly explain the tempering process. (3)

3.3.3 Name TWO disadvantages of the tempering process. (2)

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QUESTION 4

4.1 Complete the following sentences by filling in the missing word or words from the list given below. Write only the word or words next to the question number (4.1.1– 4.1.5) in the ANSWER BOOK.

The Brinell hardness test is performed as follows:

optically; dwell; staying; 100 to 150; indenter; dividing; 10 to 15; manually; multiplying; tester

4.1.1 The … is forced into sample by accurately controlled test force.

4.1.2 The force is maintained for a specified time called the … time.

4.1.3 The specified time in QUESTION 4.1.2 is between … seconds.

4.1.4 The size of the forced mark is determined … by using a portable microscope.

4.1.5 The Brinell hardness number is calculated by … the test force by the curved surface area of the mark created.

(5 × 1) (5)

4.2 The formula for calculating stress in a material is: $\sigma = \frac{F}{A}$. What are the SI units of the stress (σ), force (F) and cross-sectional area (A)? (3)

4.3 A Brinell hardness tester was used to test the hardness of a steel test piece using a 10 mm diameter steel ball. The $\frac{F}{D^2}$ value was 30. The diameter of the impression was 4,17 mm.

Calculate the following:

4.3.1 The load applied to the ball in kg. (2)

4.3.2 The surface area of the impression accurate to three decimal places. (4)

4.3.3 The Brinell hardness number accurate to three decimal places. (3)

4.3.4 The tensile strength of the hardened test piece accurate to two decimal places. (2)

Given:

$$Surface\ area\ of\ impression = \frac{\pi D}{2}\left\{D - \sqrt{D^2 - d^2}\right\}$$

$$BHN = \frac{load\ kg}{curved\ surface\ area\ of\ indentation\ (mm^2)}$$

$$Tensile\ strength = 3{,}24 \times BHN$$

4.4 A force of 16 000 N is applied to a circular work piece inducing a stress of 47,23 MPa in the work piece. Calculate the diameter of the work piece in mm accurate to two decimal places. (6)

Given:

$\sigma = \frac{F}{A}$ *and* $A = \frac{\pi}{4}D^2$

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TOTAL: 100

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